First-Principles pH Modeling for an Inline Acetate Buffer System

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pH Modeling and Control Update

Acetic acid – sodium acetate – Arium water inline mixing

Presentation Outline

1. Why Henderson-Hasselbalch is the first-principles baseline.
2. Why water is weak as a direct steady-state pH input.
3. What the updated data show.
4. Where the model prediction becomes biased.
5. How apparent pKa was calculated and interpreted.
6. Controller structures from the source document.

The Acetic-Acid/Acetate Pair Is the Buffer

Source-document terminology

- Flow 0: acetic acid, F_H
- Flow 1: sodium acetate, F_A
- Flow 2: water, F_W
- Acid and acetate stocks: about 100 mM

Henderson-Hasselbalch baseline

$$\mathrm{pH} \approx pK_a + \log_{10} \frac{C_{A^-}}{C_{HA}}$$

$$C_{HA} = \frac{C_0 F_H}{F_H + F_A + F_W}, \qquad C_{A^-} = \frac{C_0 F_A}{F_H + F_A + F_W}$$

$$\mathrm{pH} \approx pK_a + \log_{10} \frac{F_A}{F_H}$$

Main modeling split

Separate the **chemical pH effect** from the **hydraulic/dynamic effect**.

The actual pH actuator is mainly the sodium-acetate/acetic-acid ratio.

Water Cancels From Ideal pH, But Not From Dynamics

Why water is weak for steady pH

In the acetate-buffer region, water is a weak steady-state pH input. The model should not force water to explain pH changes.

What the data check found

- Charge-balance model includes dilution by water.
- Difference from HH: about 0.001 pH on average.
- Maximum checked difference: less than 0.009 pH.

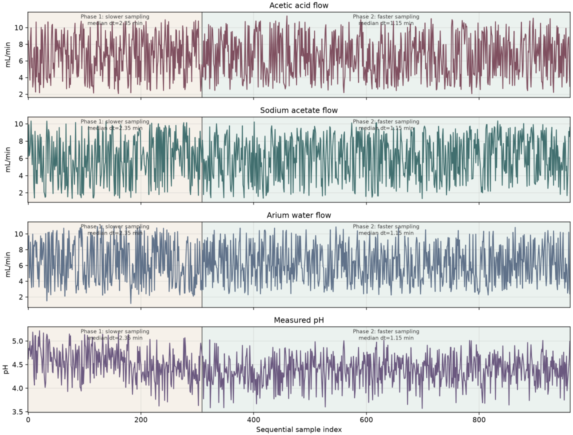
How water still matters

$$C_{\mathrm{buf}} = \frac{C_0(F_H + F_A)}{F_T}, \qquad F_T = F_H + F_A + F_W$$

$$\tau \approx \frac{V_{\mathrm{eff}}}{F_T}$$

- dilution and buffer strength
- total flow and residence time
- mixing delay and PH_2 sensor response
- activity coefficients and apparent pKa

The Updated Data Are Treated as One Sequential Record



method=data_preparation | run_time=2026-06-24 12:39:26

Prepared features

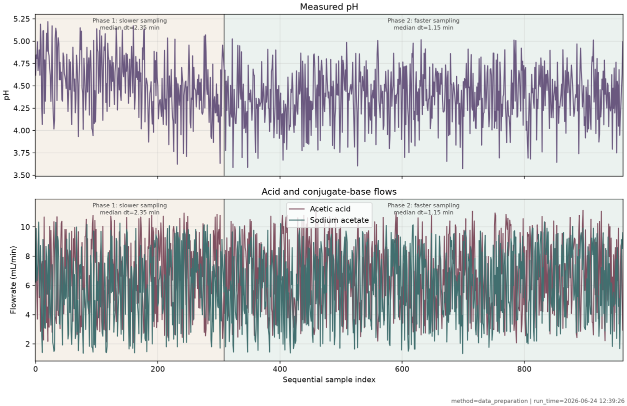
- timestep column: `time`
- last four columns:
- `flow-acid`
- `flow-sodium`
- `flow-water`
- `pH-sensor`

Sampling phases

phase	samples	median Δt
1	0–308	2.347 min
2	309–961	1.152 min

The x-axis is sample index, so long calendar gaps do not create empty plot regions.

PH_2 and the Last Column Are the Reliable pH Source

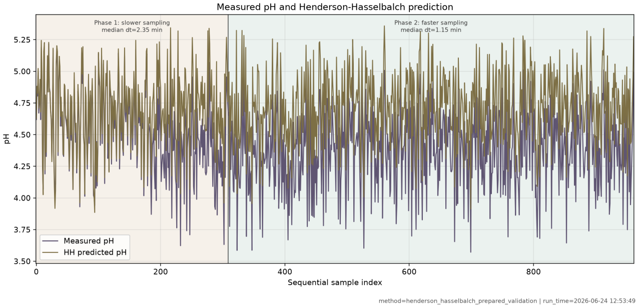


Sensor choice

- Model validation uses the last-column `pH-sensor`.
- `pH-sensor` matches PH_2.
- 962 rows compared.
- Maximum absolute difference: about 5×10^{-10}.
- PH_1 is not used for pH metrics.

The validation target is PH_2 / pH-sensor.

The HH Model Follows the Trend But Has a Bias



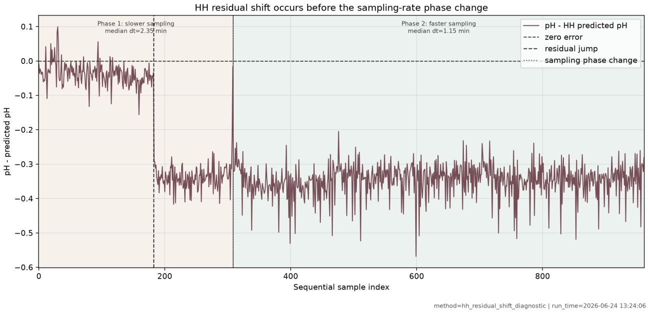
Overall metrics

metric	value
n	962
mean error	-0.286
MAE	0.287
RMSE	0.314
max abs. error	0.568
correlation	0.913

Interpretation

The acid/base ratio explains the direction of pH movement, but the static model misses an intercept-like effect.

The Residual Shift Starts at Sample 183, Not 309



Changepoint evidence

- Bias starts at sample 183.
- Sampling-rate phase change starts at sample 309.
- Sample 183 follows an overnight/session gap.
- Reservoir mass readings reset upward.

Mean residual

segment	residual
0–182	-0.037
183–308	-0.336
309–961	-0.347

Apparent pKa Is a Lumped Intercept Diagnostic

Calculation

$$pK_{a,\mathrm{app},k} = \mathrm{pH}_k - \log_{10}\left(\frac{C_{acetate,0}F_{acetate,k}}{C_{acid,0}F_{acid,k}}\right)$$

$$C_{acid,0} = C_{acetate,0} \quad \Rightarrow \quad pK_{a,\mathrm{app},k} = \mathrm{pH}_k - \log$$

What changed in the data

segment	residual	$pK_{a,\mathrm{app}}$
0–182	-0.037	4.723
183–308	-0.336	4.424
309–961	-0.347	4.413

Not the pure textbook pKa

It includes sensor bias, activity effect, temperature effect, stock concentration error, and mixing/delay error.

Before sample 183 looks reasonable; after sample 183 is a different intercept regime.

The Post-183 Apparent pKa Is Not a Temperature pKa

Temperature explanation check

- Literature pKa near room temperature is about 4.76.
- The source document notes normal aqueous temperature dependence stays near this range.
- The post-183 value is about 4.41.
- This is too large to be normal temperature variation.

Equivalent ratio interpretation

$$10^{-0.336} \approx 0.46$$

After sample 183, the data behave as if the effective acetate/acid strength ratio were about half of the assumed value, or as if the reliable pH sensor had an offset of about -0.3 pH.

Interpret this as a session-dependent calibration/intercept shift.

Raw Three-Flow Control Is Not a Unique pH Problem

Direct raw action

$$u = \begin{bmatrix} F_{acid} \\ F_{acetate} \end{bmatrix}$$

$$\mathrm{pH} \approx pK_a + \log_{10} \frac{F_{acetate}}{F_{acid}}$$

Same pH, different usage

$$(1,1), \quad (3,3), \quad (8,8) \quad \Rightarrow \quad \frac{F_{acetate}}{F_{acid}} = 1$$

Document warning

- The pH-tracking reward is not uniquely defined.
- RL can drift to unnecessarily high flows.
- RL can drift to very low flows.
- Pump actions can chatter.
- Water or total flow can look useful for a pH problem.

Do not let RL freely choose acid and acetate independently for the first controller.

Controller Structure: Ratio-Based Controller

Reparameterize the action

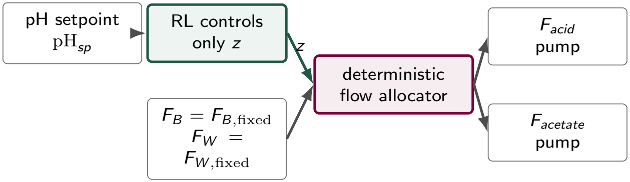
$$r = \frac{F_{acetate}}{F_{acid}}, \qquad z = \log_{10} \frac{F_{acetate}}{F_{acid}}$$

$$\mathrm{pH} \approx pK_a + z$$

$$F_B = F_{acid} + F_{acetate}$$

Deterministic flow allocator

$$r = 10^z, \quad F_{acid} = \frac{F_B}{1+r}, \quad F_{acetate} = \frac{rF_B}{1+r}$$



First RL version: RL controls only z; F_B and F_W are fixed or chosen by a simple rule.

Controller Structure: HH Baseline + RL Correction

Baseline plus correction

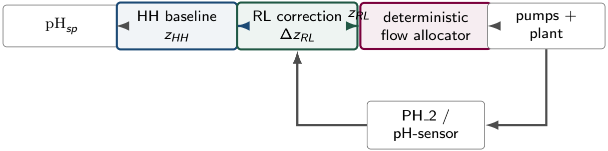
$$z_{HH} = \mathrm{pH}_{sp} - pK_a$$

$$z_{RL} = z_{HH} + \Delta z_{RL}$$

$$r = 10^{z_{RL}}$$

What Δz_{RL} corrects

- sensor bias
- mixing delay
- activity effects
- model mismatch



This is more physically meaningful than asking RL to discover the acid/base ratio from raw flows.

Controller Structures to Carry Forward

1. Ratio-based controller

- $a_k = \Delta z_k$ or $a_k = z_k$
- deterministic flow allocator
- removes non-uniqueness

2. HH baseline + RL correction

- $z_{HH} = \mathrm{pH}_{sp} - pK_a$
- $z_{RL} = z_{HH} + \Delta z_{RL}$
- correction learns mismatch

3. Chemical usage objective

- allow RL to control F_B
- reward must include usage objective
- require $C_{buffer} \geq C_{min}$

Clean hierarchy from the document

pH tracking	control acid/acetate ratio
buffer strength	control total acid+acetate flow
dilution/residence time	control water or total flow

Thank you